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OHM'S LAW

1 Theory

Electric circuits are essentially combinations of electrical devices that provide a pathway, potentially enabling charges to move. To understand how this movement happens, we rely on a fundamental principle known as Ohm's law. Ohm's law states that the loss of energy per charge, or potential difference (V), dictates how electricity behaves in a conductor, represented by the formula:

$$V = IR \tag{1}$$

where:

- V is the electric potential difference, or voltage (in volts, V)
- I is the electric current (in amperes, A)
- R is the electrical resistance (in ohms, Ω)

To make voltage and current easily distinguishable, it helps to think of an electrical circuit like a plumbing system. The electric current, represented by I , is the actual physical flow of water moving through the pipes. In a circuit, current is the flow of charged particles, like electrons. Mathematically, it is defined as the rate of charge flow over time, given by the formula:

$$I = \frac{q}{t} \tag{2}$$

where:

- I is the electric current (in amperes, A)
- q is the amount of flowing charge (in coulombs, C)

- t is the elapsed time (in seconds, s)

However, for water to flow through a pipe, it needs a push. In our analogy, that push comes from a water pump, which acts just like a battery in an electric circuit. The battery provides an electric potential difference, or voltage (V). Voltage is the energy per charge, measured in units of volts. Positively charged particles naturally want to move from an area of higher electric potential to a lower electric potential.

Finally, we have the pipes themselves. If a pipe is incredibly narrow, the water will have a hard time getting through. In a circuit, this bottleneck is called resistance (R), measured in ohms (Ω). Resistors are simply materials that decrease the flow of charges by dissipating the energy in the form of heat.

By understanding Ohm's law, we can fully justify our experimentation. Equation 2 ($V = IR$) tells us that an electric potential difference (V) will drive an electric current I given by V/R . Therefore, if we know the voltage pushing the charges and the current flowing through, we can make a prediction of the resistance. This lab tests whether these theoretical relationships hold up in real, physical circuits when we arrange carbon resistors in different ways, such as series or side-by-side in parallel.

2 Conclusion

Activity 1: Carbon Resistors

On Activity 1, our team explored the properties of carbon resistors by first identifying them using their standard color codes and then physically measuring their resistance using an ohmmeter. We measured four resistors with values of 326Ω , 985Ω , 10020Ω , and 21600Ω , which showed an extremely low percent difference from the predicted color code values—specifically 1.21%, 1.50%, 0.20%, and 1.82% respectively. Furthermore, when we mounted the resistors in series, we observed that their total resistance accurately summed up to match our predictions, yielding 1330Ω for two resistors (only a 1.44% error) and an impressive 33000Ω for all

four together (just a 0.21% error from the expected $32,931\Omega$). When we mounted them in parallel, the combined resistance appropriately dropped lower than the smallest individual resistor, measuring 6800Ω and 243Ω for two different pairs, closely tracking the theoretical inverse addition predictions of 6844.8Ω and 244.9Ω .

Activity 2: Ohm's Law

For Activity 2, we directly tested Ohm's law by placing our resistors in series with a direct current (DC) power supply. We utilized a multimeter as a voltmeter to measure the 6V potential difference across the resistors and as an ammeter to measure the electric current flowing through them. After analyzing the data, our team measured voltage-to-current slope values of approximately 2206, 327.9, 566.0, and 1000 V/A for the individual components. Furthermore, when testing the complete configuration, we measured an overall system value of 897Ω , which cleanly correlates to the calculated theoretical resistance of the circuit (896.8Ω). Overall, we found an a consistent correlation between our experimental data and the theoretical predictions, making it possible to safely validate Ohm's law and establish the rules for how resistance adds up in both series and parallel configurations.

3 Questions

Activity 1 Questions

1. **Compare the measured values of the four carbon resistors with their color code values and discuss the percent difference.**

The four selected resistors had expected color-code values of 330Ω , 1000Ω , 10000Ω , and 22000Ω . When measured with an ohmmeter, the actual resistances were 326Ω , 985Ω , 10020Ω , and 21600Ω . The percent differences were exceedingly low b1.21%, 1.50%, 0.20%, and 1.82%, respectively, demonstrating that the color-code bands provide a highly reliable estimation of a carbon resistor's true value.

2. **Infer a simple rule for resistors in series. What is the expected value of the four resistors all mounted in series, and how does it compare to the actual measurement?**

Based on the initial measurements, when resistors are mounted head-to-tail in series, their equivalent resistance is simply the sum of their individual resistances. Using this additive rule, the expected total resistance for all four components in series is $32931\ \Omega$. The physical measurement yielded $33000\ \Omega$, which is only a 0.21% error, firmly supporting the series addition rule.

3. **How is the new resistance of a parallel circuit related to the individual values? Predict the resistance of the first and second resistors mounted in parallel and compare with the actual measurement.**

For resistors mounted in parallel, the inverse of the total equivalent resistance is equal to the sum of the inverses of each individual resistance. Using this relationship, the expected parallel resistance of R_1 ($326\ \Omega$) and R_2 ($985\ \Omega$) was calculated to be $244.9\ \Omega$. The actual ohmmeter measurement was $243\ \Omega$, confirming that parallel configurations decrease the overall resistance of the pathway.

Activity 2 Questions

1. **From your results, infer four general rules about voltages and currents in series and parallel configurations.**

Based on the circuit behavior, the four established rules are: (1) In a series circuit, the electric current (I) remains constant across all components. (2) In a series circuit, the total voltage (V) is the sum of the individual voltage drops. (3) In a parallel circuit, the voltage (V) remains constant across all branches. (4) In a parallel circuit, the total electric current (I) is the sum of the currents flowing through each individual branch.

2. **Using Ohm's law, measure the total resistance between points a and d.**

Explain your method.

To calculate the total resistance using Ohm's law ($R = V/I$), the total electric potential difference supplied by the DC battery (6 V) was divided by the total electric current entering the circuit at point "a". This mathematical approach treats the entire complex circuit as a single equivalent resistor.

3. Find the expected equivalent resistance between a and d using the established rules, and compare it with the multimeter measurement.

Using the series and parallel addition rules established in Activity 1, the theoretical equivalent resistance for the entire circuit was calculated to be $896.8\ \Omega$. When the physical circuit was tested with a multimeter, the measured total resistance between "a" and "d" was exactly $897\ \Omega$. This near-perfect alignment validates both Ohm's law and the theoretical rules for circuit analysis.

4 Above and Beyond

While building simple circuits on a lab bench might seem like an abstract exercise, the principles behind Ohm's law form the foundation of the modern digital world. When we investigate how resistors limit current and drop voltage, we aren't just looking at the dissipation of heat; at a deeper level we are looking at a form of control.

Historically, early scientists and engineers realized that by carefully arranging electrical components like resistors, they could dictate exactly how electricity flows through a system. This ability to easily manipulate voltage levels became a powerful way to represent information. A high voltage could signify a 1, and a low voltage could signify a 0.

By arranging these electrical components into deliberate patterns, engineers and scientists discovered how to construct physical logic gates. This was a revolutionary breakthrough because it established a direct connection between electrical behavior and Boolean algebra. It is incredibly profound to realize that the exact same natural phenomenon that causes

a tiny carbon resistor to warm up during our lab experiment is the very mechanism that, when scaled into microprocessors, allows us to calculate satellite trajectories, render incredible graphics, and even process the text of this document. Nature provided us with the predictable relationship between voltage, current, and resistance; and the human ingenuity that harnessed that relationship to construct the computing machines we take for granted today.